

Recessed Lighting in the Limelight

Recessed downlights are among the most popular installed lighting fixtures for new and remodeled homes. One study in the northeastern United States found that new homes had an average of 23 recessed cans per home. Pacific Northwest National Laboratory (PNNL) estimates at least 350 million recessed downlights are currently installed in U.S. homes, and around 20 million are sold each year.

All those replicating fixtures seem to offer irresistible efficiency opportunities. But the vast majority of recessed downlights use incandescent light sources, because incandescent reflector lamps are readily available in various wattages and sizes and are relatively inexpensive. In California, 60% of homes built since 1990 contained recessed downlights, according to a recent study, but only 0.4% of these fixtures used compact fluorescent lamps. At the national level, annual reported sales of fluorescent residential recessed downlights make up less than 1% of total residential recessed downlight sales.

Standard recessed downlights, which often allow air to leak from conditioned to unconditioned space, waste both lighting and space conditioning energy (see Figure 1). (See also "A Recessed Can of Worms," *HE* Jan/Feb '01, p. 42). Many states are addressing the air leakage problem through building energy codes. To date, 33 states have adopted building codes that require recessed cans installed in the building shell to be airtight. (For a complete description of the air leakage standard, reference test procedure ASTM E283. ASTM E-283 is an air leakage standard for window and door assemblies that was subsequently adopted in the 1995 Model Energy Code as a way to test and quantify the reduced air leakage requirement for recessed cans.) California recently adopted this requirement in its revised Title 24 building energy standards, which will become effective in October 2005. As California's code already requires lighting fixtures in kitchens and bathrooms to use a high-efficiency light source, the market for airtight CFL recessed cans is expected to grow in the nation's most populous state.

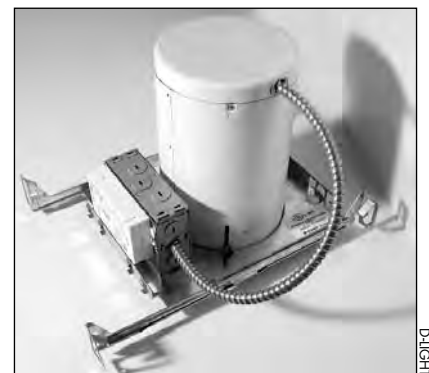


Surmounting Technical Challenges

Still, the transition to full deployment of high-efficiency recessed downlights is not a seamless one. These lights pose unique operating challenges. Temperature extremes can affect both the light output and the life span of a CFL. In an airtight recessed can installed in insulation, heat has little opportunity to dissipate, causing the temperature inside the can to rise. PNNL tested 11 recessed downlights in a simulated insulated ceiling, and found that measured ballast case temperatures ranged from 69.1°C to 110.0°C, compared to the industry average maximum rated temperature, which is about 75°C.

To encourage lighting fixture manufacturers to develop and bring to market high-efficiency recessed cans that perform well in airtight insulated ceilings, the DOE is sponsoring the Recessed Downlights project. As a part of this project, PNNL developed a specification for high-efficiency recessed downlights, including the following requirements:

- The fixture must use pin-based CFLs—to prevent replacement with incandescents—with average rated life of 10,000 hours.



- The housing of the fixture must be airtight to prevent air leakage into unconditioned attics or other spaces.
- The fixture must be IC rated for direct contact with insulation.
- The fixture must have electronic ballasts for quiet, flicker-free operation.
- At least 50% of the lamp's light output must leave the fixture, which translates into a fixture efficiency of at least 50%.
- The lamp must have a minimum light output of 900 initial lumens, net of fixture losses, to approximate the sustained light output of a 65W Energy Policy Act of 1992 (EPACT) reflector lamp. This specification takes into account the typical lumen depreciation of 20%–40% over the life of the CFL.

In addition, PNNL developed a test regime to evaluate the performance of these fixtures. The purpose of the tests was to assess the reliability of the fixtures, and to determine their light output and operating temperature when operated in an insulated ceiling. To simulate an insulated ceiling, fixtures were placed in an enclosure filled with 12 inches of loose-fill cellulose insulation

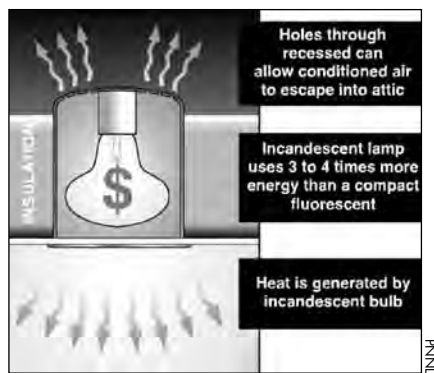


Figure 1. Lighting and conditioning energy loss through uninsulated recessed can lights can be considerable—especially when an incandescent bulb is involved.

is very important that the ballast case temperature remain below its rated maximum, and the four fixtures described below succeeded in doing so (see Table 1). Other fixtures exceeded their rated ballast case temperature during the testing period, thus failing the test.

Several new energy-efficient recessed downlights that performed well in both the long-term and short-term tests are now available at specially negotiated prices (see Table 2). To take advantage of these prices, contact the manufacturer directly and reference the DOE Recessed Downlights project.

multifamily housing. TCP's retrofit downlights were recently featured in a tour of remodeled homes in the Northwest. TCP also provides a full line of CFLs and other high-efficiency lighting fixtures, including surface mounts, table lamps, and floor lamps.

D-Light, a division of DiSci Labs LLC, is based in Corpus Christi, Texas, and has manufacturing facilities in Sparks, Nevada. They offer high-quality recessed downlights for new construction applications. Two of their new construction models have successfully completed laboratory testing conducted by PNNL.

Table 1. Ballast Case Temperature Testing

Company	Model Number	Rated Maximum Ballast Case Temp (°C)	Average Tested Ballast Case Temp (°C)
D-Light	8545	85	73.3
D-Light	8546	85	70.1
PowerLux	PI20N26	70	58.8
TCP	I3726FX	71	64.1

Table 2. Fixtures That Passed PNNL Testing

Brand	Model	Wattage	Fixture Type
D-Light	8545	26	New construction
D-Light	8546	26	New construction
PowerLux	PI20N26	26	Retrofit kit or new construction
TCP	I3726FX	26	New construction
TCP	I3726	26	Retrofit kit
TCP	I3426 Universal	26	Retrofit kit

(equivalent to about R-45). The fixtures were subjected to short-term and long-term tests within this environment.

In the short-term test, fixtures were subjected to continuous operation for 12 hours. Temperature measurements were taken at the ballast case, the lamp base, and the ambient air surrounding the lamp. The temperature of the ballast case had to remain below the rated and warranted maximum for the fixture to pass the test. Room temperature during the test was held at $25^{\circ}\text{C} \pm 1^{\circ}\text{C}$.

In the long-term test, fixtures were subjected to cyclic operation (three hours on, 20 minutes off) for 12 months. Lumen output measurements were recorded every other week. To pass this test, the fixtures had to operate without failure for the duration of the testing period. This test approximates eight years of normal operation at three hours per day.

One of the key parameters tested by PNNL was ballast case temperature, which greatly affects the life of the CFL. For every 10°C increase in ballast case temperature, ballast life is halved. It

Innovative Lighting

PowerLux Corporation of Carlsbad, California, has developed an innovative recessed downlight instant retrofit kit that addresses the problem of heat buildup by placing the electronic ballast in the trim ring of the fixture, below the ceiling and away from destructive lamp heat. This patented feature is called the PowerRim. PNNL testing has shown that the PowerRim fixture operates at substantially lower temperatures than fixtures where the ballast is located in the insulated ceiling. The PowerRim fixture is UL listed for both retrofit and new construction applications. The PowerRim has been installed in retrofit and new construction projects in hotels, theme parks, schools, and commercial and residential properties.

Technical Consumer Products (TCP), Incorporated, of Aurora, Ohio, offers energy-efficient recessed downlights in both retrofit and new construction models. TCP products have been featured in Energy Star homes, as well as in hotels and

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Recessed Downlight Manufacturers

PowerLux Corporation
Carlsbad, California
Web site: www.Powerlux.com

Technical Consumer Products (TCP), Incorporated
Aurora, Ohio
Web site: www.tcpi.com

D-Light
DiSci Labs LLC
Corpus Christi, Texas
Web site: www.downlight.net